NodeJS

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Agenda

01 Events

02 Steps in Writing Event Driven Code

03 EventEmitter

04 Registering Callbacks

05 Emitting Events

06 Emit and Respond to Events

07 Benefits of Event Driven Programming

08 Streams

09 Types of Streams

10 Read Data from a file using streams

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Agenda

11 Write Data to a file using streams

12 Stream Events

13 Process Stream Events

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MongoDB

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What is MongoDB?

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What is ExpressJS?

MongoDB is Document Oriented, NoSql database management system

Since MongoDB is a NoSql Database, you do not need to provide a schema and can insert data of any shape in a document

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Why MongoDB?

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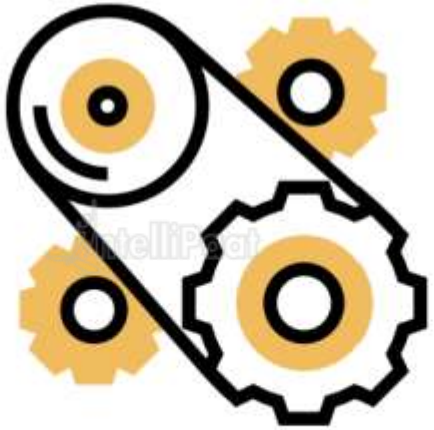
Why MongoDB?

MongoDB provides several benefits

- Fast
- Flexible
- No Joins
- Ease of Use
- Scalable

MongoDB provides high speed reads and writes to collections and documents



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Why MongoDB?

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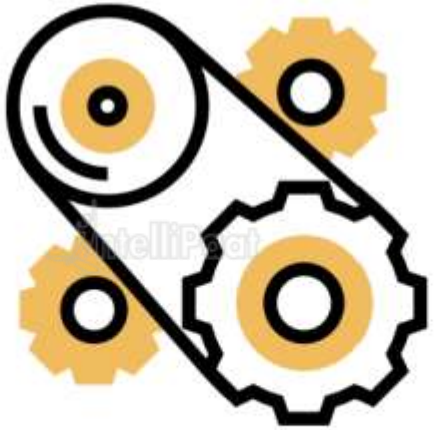
Since MongoDB is Schema Less it provides you with the flexibility of inserting a record with any shape



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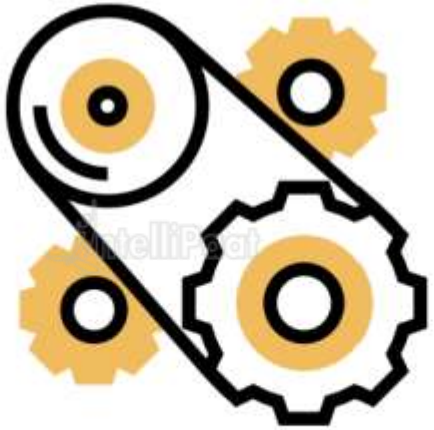
In MongoDB we can nest collections so we don't have to perform complex join queries to get results



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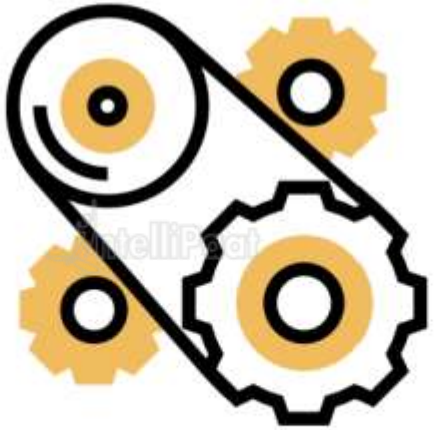
MongoDB uses query language that is similar to JavaScript and is very easy to understand and use



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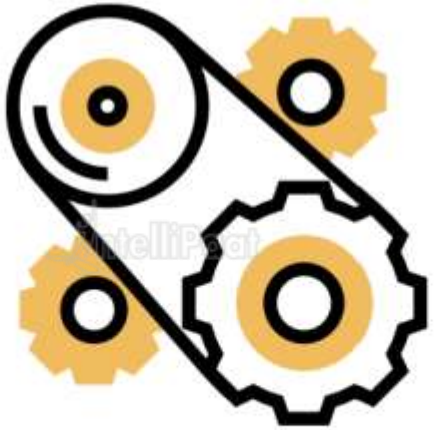
MongoDB is very easy to scale horizontally i.e. distributing data across a cluster of large number of server



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Document

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Document

A Document in MongoDB is a record inside a collection

Data is Stored inside these documents in a JSON like format called BSON or Binary JavaScript Object Notation

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Collections

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Collections

In MongoDB Database data is stored in documents and these documents are grouped inside a collection

You can think of a document as a row and a collection as a table in an SQL database

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MongoDB Data Types

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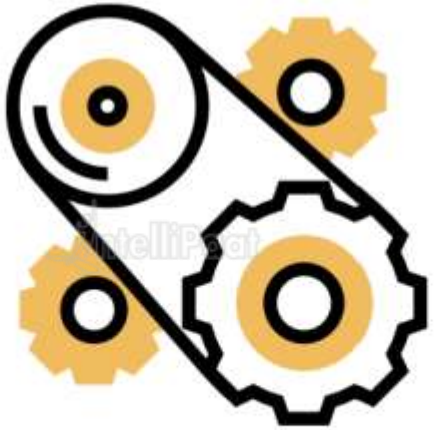
MongoDB Data Types

MongoDB supports all the data types that JSON supports with some additional data types, some of them are:

- Integer
- Double
- Timestamp
- Binary data
- Object ID
- Null

Integers are used to store whole number, depending on your server these can be 32 bit or 64 bit



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MongoDB Data Types

MongoDB supports all the data types that JSON supports with some additional data types, such as:

- Integer
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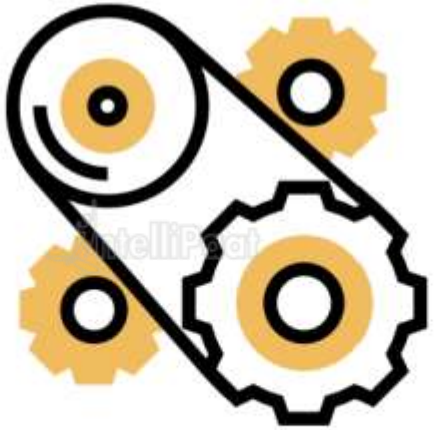
Doubles are used to store a floating point number



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MongoDB Data Types

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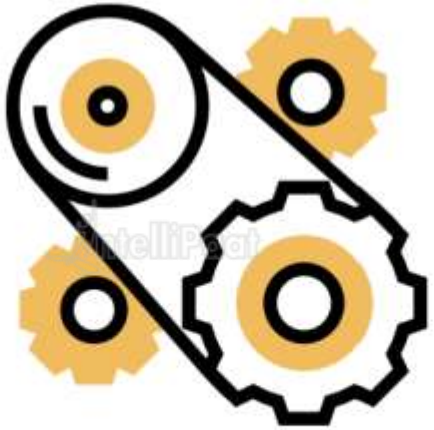
- Null

Timestamps are used to record when a document has been created or modified



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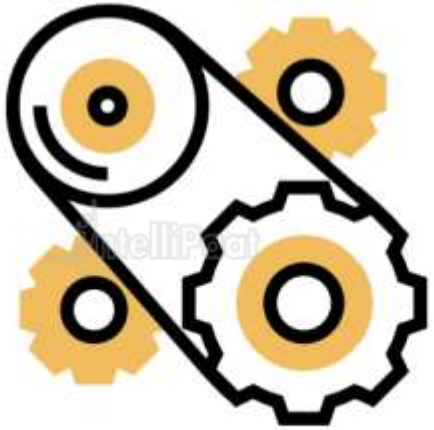
- Null

This is used to store binary data, such as images etc



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MongoDB Data Types

MongoDB supports all the data types that JSON supports with some additional data types, such as:

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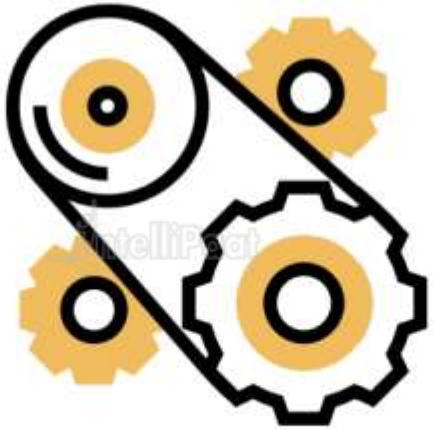
- Null

Object ID is a special type which is used to store the unique id of document



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MongoDB Data Types

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- Integer
- Double
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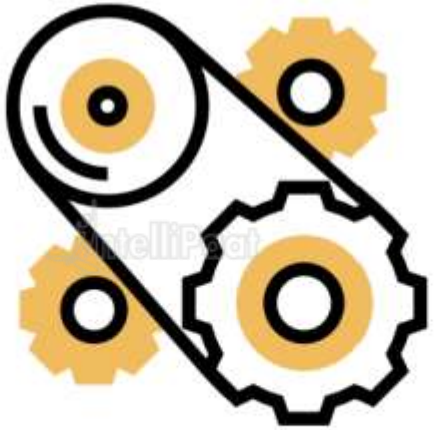
- Null

Null is data type which used to represent the absence of data



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Using MongoDB with ExpressJS

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Using MongoDB with ExpressJS

As ExpressJS is not opinionated, you can use MongoDB database with an quite easily

To use MongoDB with Express or any NodeJS application you can use one of many npm packages. Two of the most popular are the native MongoDB driver and the Mongoose package

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Native MongoDB Driver

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Native MongoDB Driver

The Native MongoDB driver is an NPM package that allows you to directly interact with your MongoDB server

It is created by the MongoDB team to allow applications to interact with the MongoDB server, very much like using the Mongo Shell

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Why use Native MongoDB Driver?

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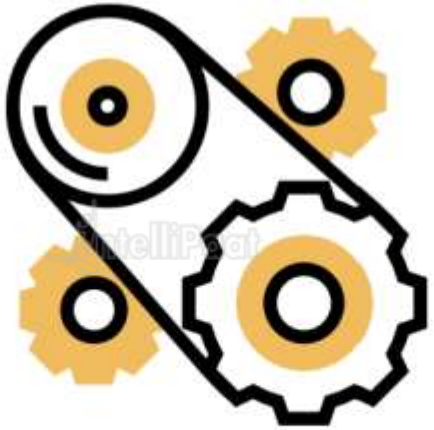
Why use Native MongoDB Driver?

There are several benefits in using Native MongoDB Driver

- Fast
- Easy
- No Schema

As it is a very thin abstraction, it provides really good performance



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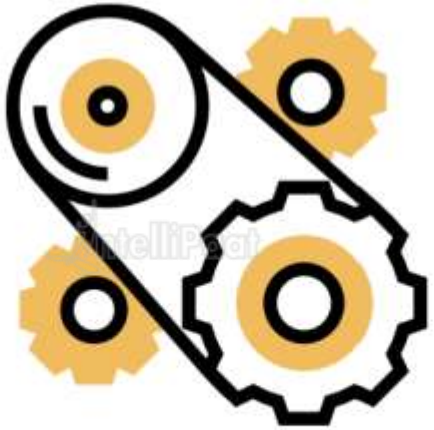
Using native driver is very easy as it make use of the same syntax that is used in the mongo shell



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Why use Native MongoDB Driver?

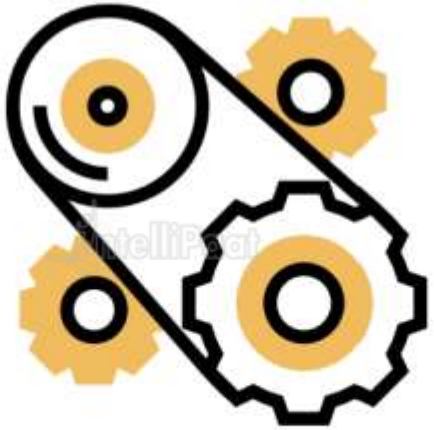
There are several benefits in using Native MongoDB Driver

Fast

Easy

No Schema

You can issue queries without using defining schema for your collections and documents



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Mongoose

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Mongoose

The Mongoose package is an ODM (Object Data Mapper)

It allows you to define schemas and models and instead of interacting with the database directly, you can just interact with the model instances and mongoose will issue the queries

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Why use Mongoose?

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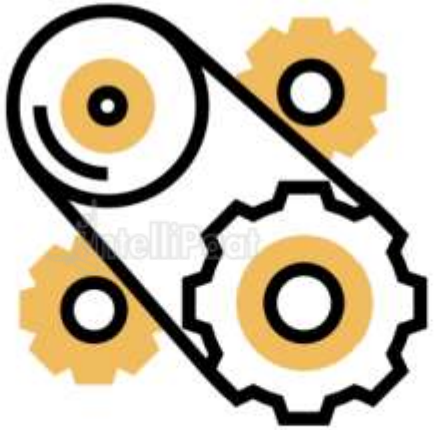
Why use Mongoose?

There are several benefits in using Native MongoDB Driver

- Speed
- Schemas
- Validation
- Utilities
- Reduced Effort

Even with all the additional benefits mongoose still provides very good data access speed



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Why use Mongoose?

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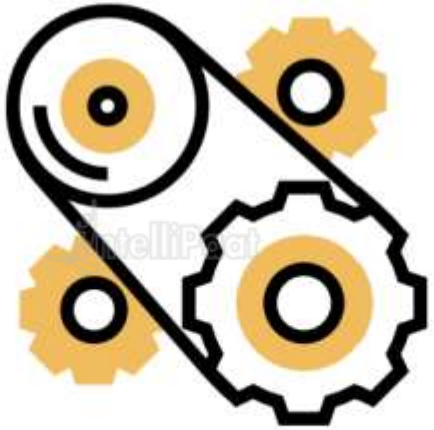
You can enforce strict schema validations to keep your data in a normalized state



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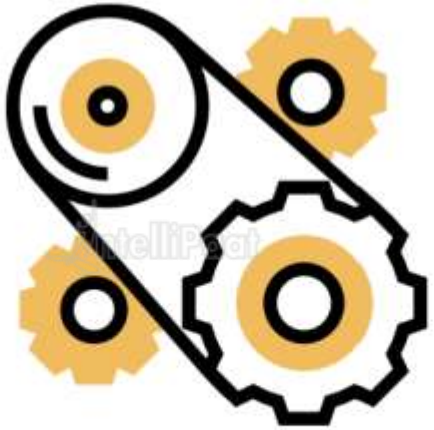
Mongoose takes validation of rules defined in schema definitions



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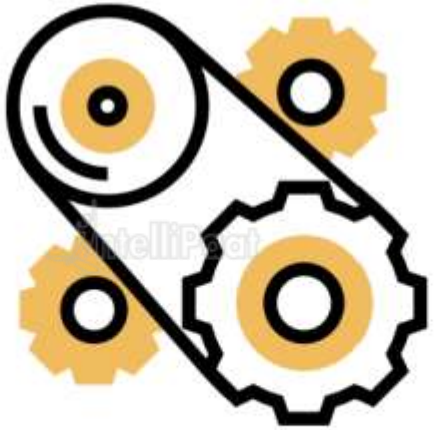
Mongoose has several built in methods that allow you to run custom code on successful query execution



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Why use Mongoose?

There are several benefits in using Native MongoDB Driver

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- Utilities
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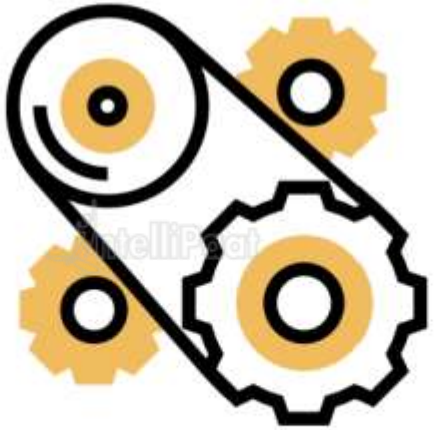
Mongoose greatly reduces the time it takes developer to build applications



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Quiz

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
D	readFile

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
D	readFile

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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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